
deep reinforcement learning for hpc job scheduling

Submission 3 — the results. Six DRL treatments, three production heuristics and a masked random control, across two real Slurm workload traces.

PRESENTER

Justin M. Cheney · 4323819

SUPERVISOR

Prof. Clement Nyirenda

INSTITUTION

University of the Western Cape

where we left off

THE LEGEND FOR EVERYTHING THAT FOLLOWS

FOUR MARKS, AND THEY HOLD FOR THE REST OF THE DECK

■ Masked DRL	MaskablePPO, MaskableDQN, MaskableA2C
□ Unmasked DRL	PPO†, DQN†, A2C†
■ Heuristic	SJF, LCFS, UNICEP
■ Masked random	The control. Same mask, same seeds, no learning

Each algorithm runs with *and* without the mask, so masking is a factor I vary rather than a choice I assume. Ten seeds per treatment, 120 runs.

AND THE TWO TRACES THEY RUN ON

Physical	87 nodes, 6,264 cores, one kind of resource. Homogeneous.
Deeplearn	30 nodes, 120 GPUs, and jobs that want different things. Heterogeneous.

READ THIS BEFORE YOU READ ANY CHART

a dagger means the run never finished an episode

Every unmasked treatment carries one. Without a mask the agent can pick a job that cannot run, the simulator advances the clock, the queue drains, and time freezes on an identical state.

Every unmasked run was stopped by a hang guard at 500,000 steps — three algorithms, ten seeds, both traces. Their numbers describe a truncated run, so they rank alongside the others without having done the same work.

the physical trace ■

RESULTS · HOMOGENEOUS WORKLOAD

FIRST: HOW MUCH ROOM WAS THERE TO IMPROVE AT ALL?

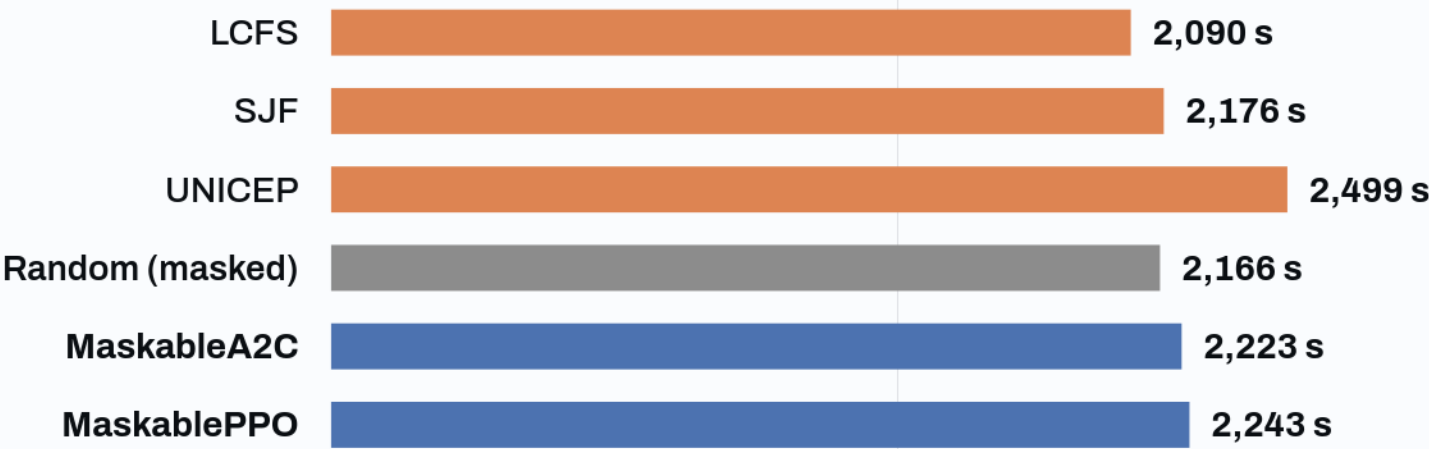
3.7%

Physical. The gap between the best heuristic and the random control. LCFS at 2,090 s against the control at 2,166 s. Almost nothing to win.

38.3%

Deeplearn. UNICEP at 846 s against the control at 1,170 s. Ten times more room for a learned policy to show a benefit.

PHYSICAL · AVERAGE WAITING TIME



■ masked DRL ■ heuristic ■ random control

0 of 6

DRL treatments beat the random control

7 of 9

comparisons favour the random control

Random guessing inside the mask is competitive here. That does not make random scheduling good.

It means the mask was doing the work, not the policy.

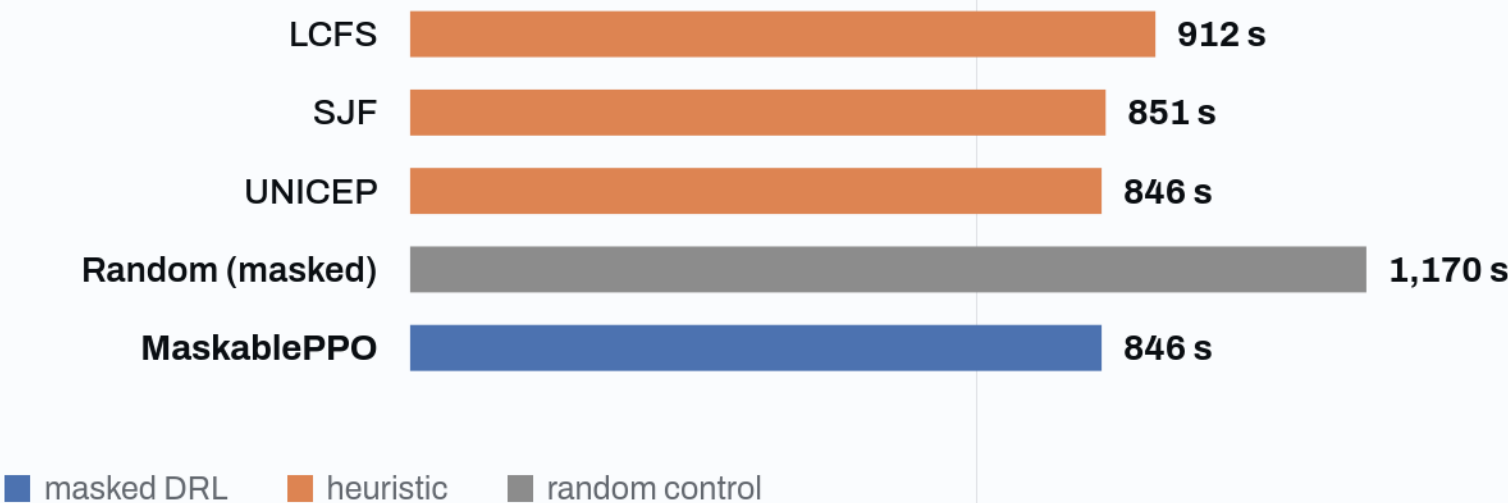
Nine comparisons in total: three masked treatments against the control, on three metrics each. Seven separate, and all seven favour random.

— On a trace with no room, nothing learned beat guessing

the deeplearn trace

RESULTS · HETEROGENEOUS WORKLOAD

DEEPLearn · AVERAGE WAITING TIME



TWO THINGS THE PANEL DOES NOT SHOW

MaskablePPO lands on 846 s, level with UNICEP. On this metric it matches the best heuristic rather than beating it.

MaskableA2C reaches a similar waiting time and still fails to separate from the control: 8 of 10 seeds, $p = 1.000$ after Holm correction. One treatment carries this trace, not the family.

10 of 10

seeds where MaskablePPO beats the control, on every primary metric

-22.0%

maximum slowdown against LCFS, a production heuristic

WHERE THE WIN ACTUALLY IS

MaskablePPO cuts **maximum** slowdown by 22.0% against LCFS and 63.2% against the control. On average waiting it sits within 1% of the heuristics.

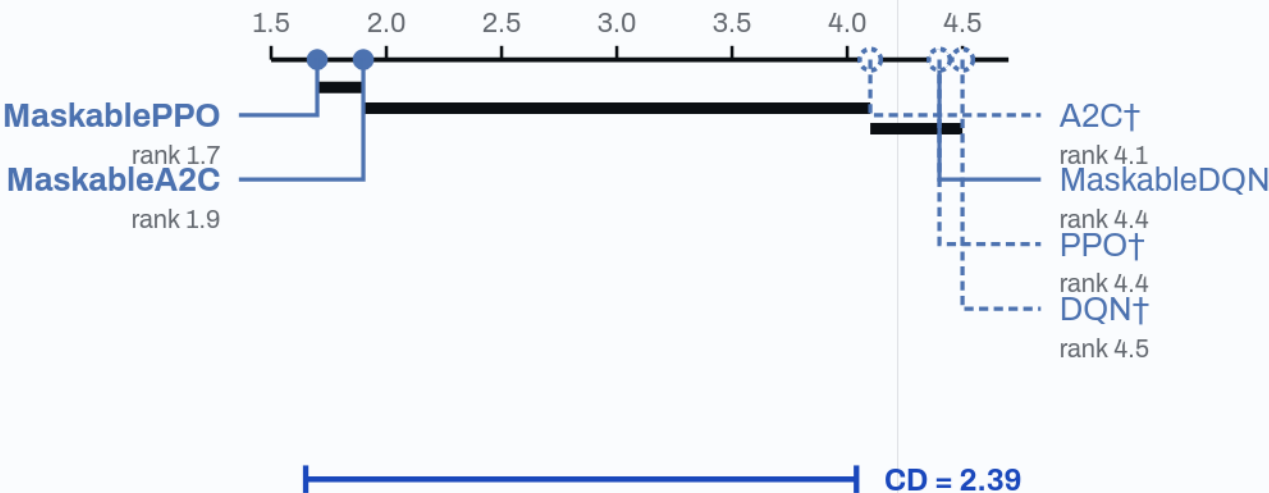
It handles the awkward jobs better, not the average one.

Awkward jobs are more common in heterogeneous workloads, which is the difference between these two traces.

and against each other ■

RESULTS · CRITICAL DIFFERENCE

PHYSICAL · MEAN RANK, AVERAGE WAITING TIME

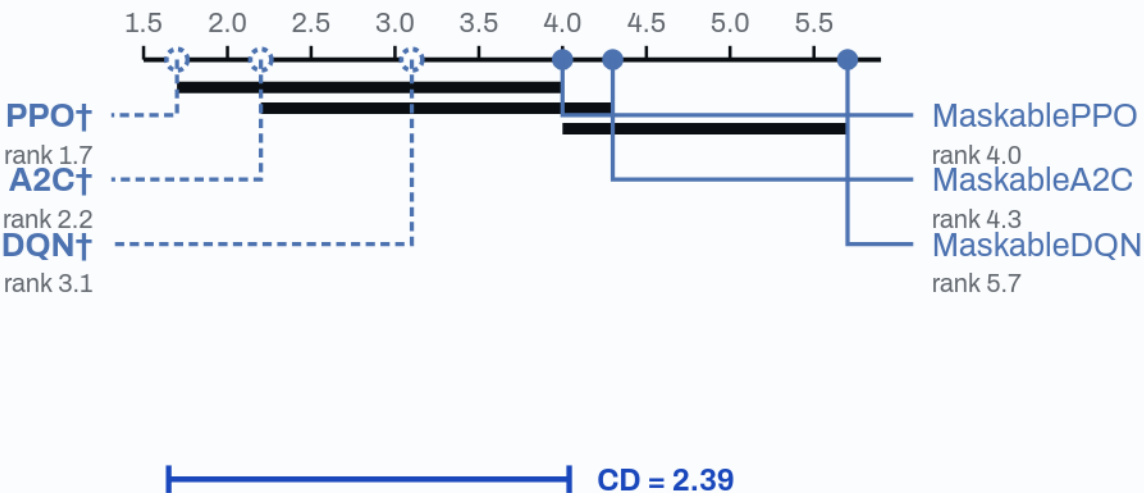


CD = 2.39 rank units at $n = 10$ and $k = 6$, which holds for both traces because the design is identical. A bar joins treatments that cannot be separated.

Deeplearn inverts the physical order, and every treatment above MaskablePPO carries a dagger. None of them finished an episode. MaskablePPO is the best ranked treatment that actually completed one.

— Six treatments, two traces, and not one clean separation

DEEPLearn · MEAN RANK, AVERAGE WAITING TIME



BETWEEN THE ALGORITHMS THEMSELVES

**nothing separates
from anything else**

THE FINDING

masking buys deployability, not
schedule quality

The deadlock shows this most directly. Without the mask an agent picks a job that cannot run, the simulator advances the clock, the queue drains, and time freezes. **Not one unmasked run finished an episode**, on either trace.

THE CLAIM I'D DEFEND

a masked random control is
cheap, decisive, and belongs in
this literature

— One cheap control changed what this study was allowed to claim

AND THE BOUND ON ALL OF IT

512

actions between reward signals

82x

gap in gradient updates, PPO vs
A2C

The reward fires once every 512 decisions, so the signal is far sparser than the horizon it is discounted over. And the same three million steps buy PPO about 7,300 gradient updates against A2C's 600,000, under inherited library defaults.

The diagnostics say the agents *optimised successfully*. They were optimising the wrong thing.

So the parity reported here is a lower bound, not a verdict on DRL.

STATISTICS

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Demšar (2006). Statistical Comparisons of Classifiers over Multiple Data Sets. *JMLR* 7, 1–30. Friedman and the CD diagram.

Holm (1979). A simple sequentially rejective multiple test procedure. *Scand. J. Stat.* 6(2), 65–70.

ALGORITHMS & MASKING

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Mnih et al. (2015; 2016). Human-level control through deep RL; Asynchronous Methods for Deep RL. *Nature* 518, 529–533; *ICML, PMLR* 48, 1928–1937. DQN and A2C.

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SYSTEMS & IMPLEMENTATION

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Yoo, Jette & Grondona (2003). SLURM: Simple Linux Utility for Resource Management. *JSSPP, LNCS* 2862, 44–60. Both traces are Slurm accounting logs.

Raffin et al. (2021). Stable-Baselines3. *JMLR* 22(268), 1–8. With sb3-contrib for the maskable variants.